

TrES-3b

R-2647

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_210607 · created 2026-08-29T22:58:02+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459372.7783 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.012
Transit depth (Rp/Rs)^2	2.22 +/- 0.35 [%]
Orbital Inclination (inc)	84.0 +/- 0.67
Airmass coefficient 1 (a1)	0.9865 +/- 0.0088
Airmass coefficient 2 (a2)	0.01 +/- 0.015
Scatter in the residuals of the lightcurve fit is	0.9 %
Best Comparison Star	#2 - [394, 421]
Optimal Aperture	4.2
Optimal Annulus	8.72
Transit Duration (day)	0.0684 +/- 0.004

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.149 $\pm$ 0.012 vs 0.16309 (deviation 1.1 σ)
Duration fit vs published	0.0684 d vs 0.0567 d (deviation 2.75 σ)
Mid-time O-C	1.3 min
Depth SNR	11.7
Residual scatter	0.9 %

Light curve

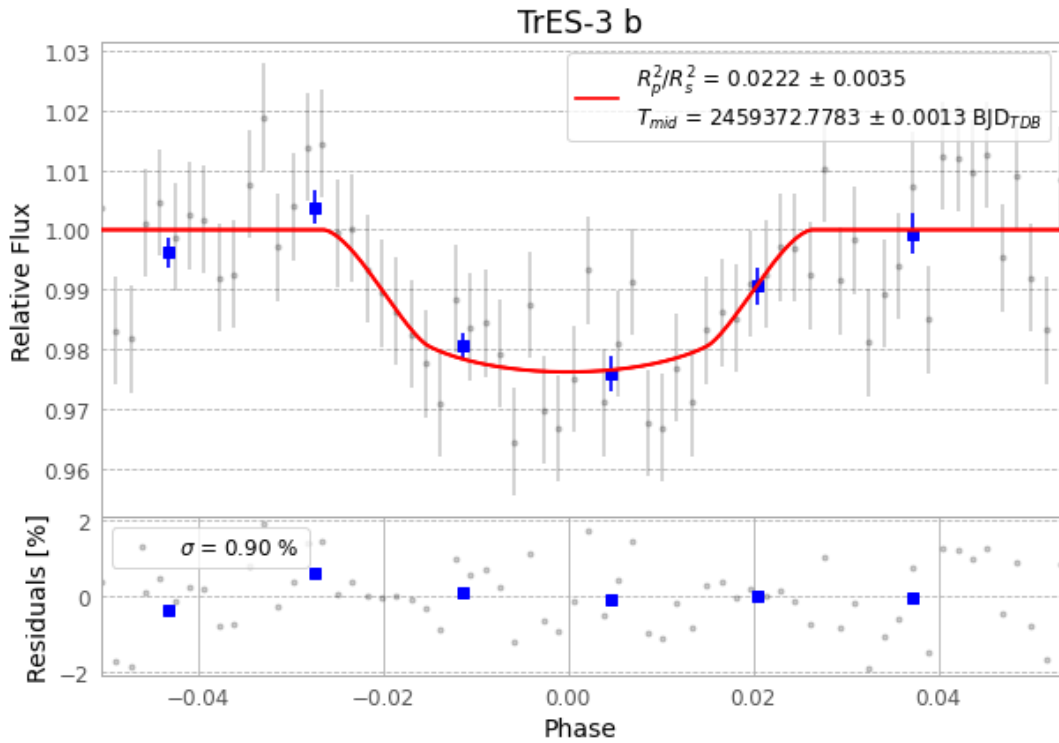


Figure 1 — FinalLightCurve_TrES-3 b_2021-06-06.png (SHA-256 7d704212cf7a9304...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [349, 207], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5ec3c2a42eadb42f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), TRES-3210607050012.FITS → TRES-3210607081512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-210607.log (9f2c16814989...), FinalLightCurve_TrES-3 b_2021-06-06.png (7d704212cf7a...), FinalLightCurve_TrES-3 b_2021-06-06.csv (7047f9f5056f...)

Compute resources

Wall time 118.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 22:58Z.